WACE Chemistry ATAR · Cheat Sheet (Patterns + Notes)

General rules: Data-booklet content (periodic table, E° table, constants, solubility rules, colours, amino acid structures, pH and ppm formulae) need not be rewritten, but know where it is · Explain with collision theory; Le Châtelier's principle only predicts · Equations: state symbols, species involved only, conditions (catalyst, acidic, heating) · Units on every value; round only at the end · No abbreviations such as NVR or NR; spell out any abbreviation at first use · No anthropomorphism (enzyme sensitive, molecules swarm, hydrophobic "hates" water) · No arrows for increase/decrease; use words.

1 Equilibrium and rate (EQ)

① Dynamic equilibrium (EQ3): Closed system: forward and reverse rates equal, macroscopic properties constant; concentrations not necessarily equal (an open system loses substances, so equilibrium cannot form).

② Le Châtelier's principle (EQ4)

- Add reactant / remove product ⇒ right; add product / remove reactant ⇒ left; solids and pure liquids do not affect equilibrium.
- Increase pressure (decrease volume) ⇒ side with fewer gas molecules; equal gas moles ⇒ no shift.
- Inert gas: constant volume ⇒ no shift; constant pressure ⇒ side with more gas molecules. ⚠ State constant volume or pressure.
- Catalyst: no shift; forward and reverse rates rise equally; K unchanged.
- Increase temperature ⇒ endothermic direction. Only temperature changes K.

③ Collision theory (EQ5): Adding reactant makes the forward rate jump; the reverse rate rises as product forms until they are equal again. Effective collision: energy ≥ Ea and correct orientation. ⚠ Explanations must use "effective collisions", not Le Châtelier.

④ Equilibrium constant (EQ7–8): K = [products]^n / [reactants]^m; pure solids and liquids excluded. Reversing the equation ⇒ reciprocal K; multiplying coefficients by n ⇒ K^n. Large K ⇒ more products; only temperature changes K.

⑤ Energy and rate graphs (EQ2, EQ6): ΔH = Ea(forward) − Ea(reverse); a catalyst lowers both Ea, ΔH unchanged. Concentration–time: no abrupt change at disturbance; rate–time: rate jumps.

⑥ Colours (EQ9): CrO_4^{2-} (yellow) ⇌ $Cr_2O_7^{2-}$ (orange): acid ⇒ deeper orange; base ⇒ yellow. Fe^{3+} (pale brown) + SCN^- ⇌ $[Fe(SCN)]^{2+}$ (blood red). $[Co(H_2O)_6]^{2+}$ (pink) ⇌ $[CoCl_4]^{2-}$ (blue): concentrated HCl or heating ⇒ blue (forward endothermic); water ⇒ pink.

⑦ Real-world contexts (EQ10): Ocean acidification: CO_2 + H_2O ⇌ H_2CO_3 ⇌ H^+ + HCO_3^-; H^+ reacts with CO_3^{2-}, shifting $CaCO_3$ formation left, so shells form less readily. Blood pH: hyperventilation (CO_2 falls) ⇒ left shift, pH rises. ⚠ Ocean pH is still above 7; shells are harder to form, not dissolved.

2 Acids and bases (AB)

① Models (AB1–2): Arrhenius: acids give H^+, bases give OH^- in water; cannot explain NH_3 as a base. Brønsted–Lowry: acid = proton donor, base = proton acceptor; conjugate pairs differ by one H^+ (HCl/Cl^-, NH_4^+/NH_3). Amphiprotic species (H_2O, HCO_3^-, HSO_4^-) donate and accept H^+. ⚠ Strong/weak = degree of ionisation; concentrated/dilute = concentration; ionisation (covalent → ions) ≠ dissociation (ions separate in an ionic crystal).

② Ionisation and Ka (AB2): Ka = $[H^+][A^-]/[HA]$; larger Ka ⇒ stronger acid. Strong acids ionise almost completely, weak acids partially. Polyprotic acids ionise stepwise: H_2SO_4 first step complete, second incomplete (HSO_4^- ⇌ H^+ + SO_4^{2-}). Weak acid solution contains HA, H_2O, H^+, A^-, OH^-; strong acid solution contains almost no HA.

③ Salt hydrolysis (AB3): Strong acid–strong base salt (NaCl) ⇒ neutral; strong acid–weak base (NH_4Cl) ⇒ acidic; weak acid–strong base (CH_3COONa, Na_2CO_3) ⇒ basic. Amphiprotic salts: $NaHCO_3$ slightly basic; $NaHSO_3$ acidic. ⚠ Use relative $[H_3O^+]$ and $[OH^-]$; do not just say "the anion hydrolyses".

④ Buffers (AB4): Weak acid + conjugate base (CH_3COOH/CH_3COO^-), or weak base + conjugate acid (NH_3/NH_4^+). Strong acid–strong base, or strong acid with its salt, is not a buffer. Add H^+: H^+ + A^- → HA; add OH^-: OH^- + HA → A^- + H_2O; the equilibrium shift cancels the change, so pH changes little. Greatest capacity at [HA] ≈ $[A^-]$, where pH ≈ pKa. ⚠ Dilution leaves pH almost unchanged but lowers buffer capacity.

⑤ pH (AB5): pH = −log$[H^+]$; at 25 °C pOH = 14 − pH (Kw = 1.00×10^{-14}). Strong base: $[OH^-]$ = base concentration ⇒ $[H^+]$ = Kw / $[OH^-]$. Decimal places of pH = significant figures of $[H^+]$.

⑥ Titration and indicators (AB6–8): Equivalence point = exact reaction; end point = indicator change; they may differ slightly. Indicators (not in booklet, memorise): methyl orange red (pH < 3.1) → yellow (> 4.4); bromothymol blue yellow (< 6.0) → blue (> 7.6); phenolphthalein colourless (< 8.2) → pink (> 10.0). The indicator range must lie within the pH jump: strong acid–weak base (equivalence pH < 7) use methyl orange; weak acid–strong base (pH > 7) use phenolphthalein; strong–strong either. ⚠ Explain the choice using the ions at equivalence (e.g. CH_3COO^- hydrolysis gives basic). Rinse burette with its own solution, pipette with its own solution, conical flask with distilled water only (rinsing with analyte adds solute; result too high). Primary standard: high purity, stable in air, large molar mass, soluble, reacts completely in stoichiometric ratio; e.g. $KHC_8H_4O_4$ (KHP), anhydrous Na_2CO_3. NaOH and HCl are not primary standards.

⑦ Non-redox reactions (AB9): Precipitates: Ag^+ + Cl^- white (AgCl); Ag^+ + Br^- or I^- pale yellow (AgBr, AgI); Ba^{2+} + SO_4^{2-} white ($BaSO_4$). Hydroxides: $Cu(OH)_2$ blue; $Fe(OH)_2$ green; $Fe(OH)_3$ reddish-brown; $Mg(OH)_2$, $Al(OH)_3$ white; $Al(OH)_3$ and $Zn(OH)_2$ dissolve in excess NaOH. Heating blue $Cu(OH)_2$ or green $CuCO_3$ turns both black (CuO).

3 Redox and electrochemistry (RX)

① Oxidation numbers (RX1): Element 0; monatomic ion = charge; compound sums to 0. O usually −2 (peroxides −1); H usually +1 (metal hydrides −1); F always −1; Group IA +1, IIA +2. Oxidation: number rises (loses e^-); reduction: number falls (gains e^-); oxidising agent is reduced, reducing agent oxidised.

② Balancing half-reactions (RX2): Acidic: balance atoms other than H and O → O with H_2O → H with H^+ → charge

with e^- → multiply so electrons match, then add. Basic: add OH^- to both sides to neutralise H^+. Multiplying a half-reaction does not multiply E°.

③ Standard electrode potentials (RX4): E° is intensive; the booklet states only 25 °C (textbook standard: 1.00 mol L^{-1}, 100 kPa). Larger E° ⇒ more easily reduced (stronger oxidant). E°cell = E°(reduction) − E°(oxidation), both written as reductions; E°cell > 0 ⇒ spontaneous. Measured voltage differs from E° due to non-standard concentration, temperature, internal resistance and overpotential.

④ Galvanic and electrolytic cells (RX5–6): Mnemonic: An Ox (anode oxidation), Red Cat (cathode reduction). Galvanic (spontaneous): anode negative, cathode positive; electrons flow anode → external circuit → cathode, never through the electrolyte; salt bridge anions move to the anode, cations to the cathode. Electrolytic (external supply, non-spontaneous): anode to positive terminal, cathode to negative; oxidation still at anode. Notation: anode | anode ions || cathode ions | cathode (single line = phase boundary; double = salt bridge). Electroplating: object is the cathode, plating metal the anode. Copper refining: impure Cu anode, pure Cu cathode; Ag and Au settle as anode mud.

⑤ Practical batteries (RX7): Alkaline zinc–manganese: primary (not rechargeable). Lead–acid: secondary; discharge consumes H_2SO_4, so density falls. Acidic H_2–O_2 fuel cell: negative $2H_2 \rightarrow 4H^+ + 4e^-$; positive $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$; only product is water, but H_2 storage, safety and cost are disadvantages.

⑥ Corrosion and protection (RX8): Iron anode: $Fe \rightarrow Fe^{2+} + 2e^-$; cathode $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; $Fe^{2+} \rightarrow Fe^{3+}$, forming rust $Fe_2O_3{\cdot}xH_2O$. Needs water and oxygen; salt accelerates. Protection: barrier (paint, grease); sacrificial anode (Zn, Mg, more reactive than Fe; jetty piles, hulls); impressed current (iron to negative terminal). ⚠ Galvanising still protects after scratching (Zn is sacrificial); tin plating accelerates corrosion once scratched.

⑦ Redox observations (RX3): Zn + $CuSO_4$: blue fades, salmon-red solid forms. Cl_2(aq) + KBr(aq): colourless turns orange. Complete combustion: blue flame, only CO_2 and H_2O; incomplete: yellow, smoky, CO or carbon.

4 Organic chemistry (ORG)

① Naming (ORG1–2): Suffixes: -ene, -ol, -al (chain end), -one, -oic acid, ester = alkyl + alkanoate, -amine, -amide (chain end). Steps: longest chain with the functional group → lowest locant → alphabetical substituents. ⚠ pentanamide (not pentamide); 2,3-dimethylbut-2-ene (all locants); ester alcohol part first (ethyl ethanoate). Show all bonds and correct atoms (HC–R, not CH–R); condensed formulae must show –COOH clearly.

② Isomerism (ORG3): Chain, positional, functional group (e.g. ethanol and dimethyl ether for C_2H_6O), cis–trans. Cis–trans needs two different groups on each alkene carbon; not all alkyl compounds have it.

③ Intermolecular forces (ORG4): Strength: hydrogen bonding > dipole–dipole > dispersion; large molecules accumulate dispersion forces, which often dominate. Boiling point: larger molar mass ⇒ stronger dispersion ⇒ higher; branching ⇒ lower; –OH, $–NH_2$, –COOH form H-bonds ⇒ higher. Solubility: solute and water must form H-bonds or dipole interactions; long chains reduce solubility; NaCl in water is ion–dipole. ⚠ Do not write only "like dissolves like"; name the forces and compare their strengths.

④ Addition, oxidation, esterification (ORG5–7): Alkene addition with Br_2, H_2, HX, H_2O (industrially H_3PO_4); bromine water orange → colourless. Acidified $K_2Cr_2O_7$ (orange → green, Cr^{3+}) or acidified $KMnO_4$ (purple → colourless or pale pink, Mn^{2+}): primary alcohol → aldehyde (distilled off as formed) → carboxylic acid (reflux, excess oxidant); secondary → ketone; tertiary: no reaction. Aldehyde → acid, also turns dichromate green. Acid + alcohol ⇌ ester + water (conc. H_2SO_4, heat, reversible); alkaline hydrolysis (saponification) is irreversible, giving a carboxylate. Acid + carbonate: CO_2 bubbles. Amines are basic (nitrogen lone pair) and form salts with acids.

⑤ Identification (ORG6, 8, 9): Write reagent + observation for each substance + conclusion (bromine water: alkene decolourises; acidified dichromate: primary/secondary alcohol turns green, tertiary unchanged; Na_2CO_3: carboxylic acid fizzes).

5 Polymers, amino acids and proteins (PRO)

① Addition polymers (PRO1): The double bond opens and the monomer carbons bond; repeat unit in brackets, n outside, extension bonds at both ends. ⚠ Do not add end groups. Monomer from polymer: remove brackets, restore the double bond.

② Condensation polymers (PRO2): Two functional groups per monomer: diol + dicarboxylic acid ⇒ polyester (–COO–, e.g. PET); diamine + dicarboxylic acid ⇒ polyamide (–CO–NH–, e.g. nylon 6,6). Each bond eliminates a small molecule (H_2O; HCl with acyl chlorides). Monomers from polymer: break C–O (ester) adding OH to the carbonyl end and H to the oxygen end; break C–N (amide) similarly.

③ Plastics (PRO3): Thermoplastic: linear or lightly branched, weaker forces, softens, recyclable. Thermosetting: covalent cross-links form a network; does not melt; hard and heat-resistant. More branching ⇒ lower density, softer (LDPE); linear, closely packed ⇒ higher density, harder (HDPE). Polyester ester bonds can be hydrolysed. ⚠ State which of chain length, branching, cross-linking or intermolecular forces is involved.

④ Amino acids (PRO4): General formula H_2N–CH(R)–COOH; all except glycine are chiral. Neutral pH: zwitterion H_3N^+–CH(R)–COO^-; acidic: cation; basic: anion. Side chains (from booklet structures): non-polar, polar uncharged, acidic (Asp, Glu), basic (Lys, Arg, His).

⑤ Peptides (PRO5): Peptide bond –CO–NH–: –COOH of one amino acid condenses with $–NH_2$ of another, losing H_2O. Dipeptide: N-terminus (free NH_2) on the left, C-terminus (free COOH) on the right; read N → C (Gly–Ala ≠ Ala–Gly).

⑥ Protein structure (PRO6): Primary: sequence joined by covalent peptide bonds. Secondary: α-helix and β-pleated sheet, held by H-bonds between backbone C=O and N–H. Tertiary: side-chain interactions: H-bonds; ionic bonds ($–NH_3^+$ and $–COO^-$); disulfide bonds (–S–S– between cysteines); dispersion forces (non-polar side chains). Denaturation (heat, strong acid or base, heavy metal ions, organic solvents) breaks H-bonds, ionic and disulfide bonds; peptide bonds remain intact. Mutation changes the primary sequence and may alter folding and function.

⚠ Do not say "heating breaks peptide bonds" or that hydrophobic groups "hate water"; say dispersion forces between non-polar side chains.

⑦ Enzymes (PRO7): Protein biocatalysts; active site complementary in shape to the substrate, highly specific; optimum temperature and pH. Industrial: mild conditions, high selectivity; limits: denature at high temperature or extreme pH, costly, hard to recover. ⚠ pH changes the shape and charge of the active site and denatures the enzyme; do not write "enzyme is sensitive to pH".

6 Synthesis and industry (SYN)

① Routes (SYN1): Alkene → alcohol (hydration); alcohol → aldehyde → acid (oxidation); acid + alcohol → ester. State reagents, conditions, intermediates and reaction type at each step.

② Haber process (SYN2): $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, exothermic; ~450 °C, ~20 000 kPa, iron catalyst; unreacted gases recycled; NH_3 cooled, liquefied and separated. Compromise: low T ⇒ higher yield, slower rate; high T ⇒ faster, lower yield; high P raises both rate and yield, but adds equipment and energy cost; catalyst only raises rate.

③ Contact process (SYN2): $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, exothermic; V_2O_5 catalyst, ~450 °C, 100–200 kPa. SO_3 is absorbed in concentrated H_2SO_4 (water gives acid mist).

④ Ethanol production (SYN1, 3): Hydration: $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$, H_3PO_4 catalyst, ~300 °C, ~6 000–7 000 kPa; petroleum feedstock (non-renewable), fast, continuous, high purity, high energy use. Fermentation: $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$, yeast enzymes, 25–37 °C, atmospheric pressure, anaerobic, mostly batch; renewable, low energy, but slow and product only ~12–15%, needing distillation. Compare routes on renewability, energy, by-products and atom economy.

⑤ Soaps and detergents (SYN4): Saponification: triglyceride + 3NaOH → glycerol + 3 fatty acid sodium salts (soap). Soap: hydrophilic head ($–COO^-\ Na^+$) + hydrophobic tail; in a micelle the tails face inwards trapping oil and the heads face outwards. Hard water: Ca^{2+} and Mg^{2+} form insoluble salts (scum) with soap anions; calcium and magnesium salts of anionic detergents are soluble, so cleaning is unaffected.

⑥ Biodiesel (SYN5): Transesterification: triglyceride + $3CH_3OH \rightleftharpoons$ glycerol + 3 fatty acid methyl esters. NaOH/KOH is fast, cheap and high-yielding but corrosive and forms soap; lipase is mild, forms no soap and can be reused (2020 key).

7 Stoichiometry and quantitative analysis (QNT)

① Conversions and dilution (QNT1): n = m/M; n = cV; dilution $c_1V_1 = c_2V_2$ (not in booklet, memorise). g L^{-1} ÷ molar mass ⇒ mol L^{-1}; ppm (mg/kg) ≈ mg L^{-1} for dilute aqueous solutions. ⚠ Write volume, concentration, mass; do not use "amount" as a substitute.

② Gases (QNT6): STP (0.0 °C, 100 kPa): molar volume 22.71 L mol^{-1} (booklet value, not 22.4 L). PV = nRT: T in K (+273.15), P in kPa, V in L; R = 8.314 used with kPa·L.

③ Limiting reagent and yield (QNT2–3): Limiting reagent = smallest n ÷ coefficient; theoretical yield is calculated from it. Yield = actual ÷ theoretical × 100%; purity = mass of pure substance ÷ mass of sample × 100%.

④ Formulae (QNT5): For each element n = m/M; divide by the smallest to get a whole-number ratio; if not near an integer, multiply by 2, 3, etc. Combustion: $n(C) = n(CO_2)$, $n(H) = 2n(H_2O)$, O by difference. Molecular formula = (molar mass ÷ empirical formula mass) × empirical formula.

⑤ Titration (QNT4): n(unknown) = n(titrant) × coefficient ratio; each mole of H_2SO_4 gives 2 mol H^+. Scaling: 25.00 mL taken from 250 mL ⇒ multiply by 10. Average concordant titres (difference ≤ 0.10 mL), discarding rough titres. Back titration: add excess reagent first, then titrate the remainder.

⑥ Precipitation gravimetry (QNT8): Precipitate completely → filter → wash with distilled water → dry to constant mass → weigh and calculate. ⚠ Incomplete drying makes the mass too high; insufficient washing makes the result too high.

⑦ Significant figures (QNT7): Multiplication/division: fewest significant figures; addition/subtraction: fewest decimal places. Round only at the end, with units; exact numbers are not counted.

8 Science inquiry (INQ)

① Design (INQ1–3): Hypothesis: testable, "If … then … because …". State independent, dependent and controlled variables and how they are controlled; include a control; repeat and average. Safety: state hazards, risks and protective measures.

② Graphing and reading (INQ4–5): Label axes with quantity and unit, even scales; title "dependent variable against independent variable". Draw a smooth line of best fit with a ruler, not point-to-point. Show read-off lines on the graph; give readings to the correct decimal places (e.g. 4.8, not 5).

③ Errors (INQ6): Random error: fluctuating readings, estimation; reduced by repeats and averaging. Systematic error: fixed-direction bias (e.g. burette not rinsed, instrument not calibrated, impure standard); repetition does not remove it. State whether the error makes the result higher or lower, and why. Reading uncertainty: half the smallest division (50 mL burette ±0.05 mL).

④ Conclusions and observations (INQ7): Conclusions must be supported by data; state limitations (small sample, large error); do not write "proves" without strong evidence. ⚠ Solutions: "colourless" or the specific colour, not "clear"; solids: colour and form, not chemical name; metals: "silver-grey solid"; copper: "salmon-red solid". Gases: "bubbles", not "colourless odourless gas" (an inference). Record all observations before, during and after the reaction, including excess or limiting reagent.

⑤ Gas tests (not in booklet): H_2: explosive pop; O_2: relights a glowing splint; CO_2: turns limewater milky; Cl_2: moist blue litmus turns red, then bleaches; NH_3: moist red litmus turns blue.